ASSISTANT ENVIRONMENTAL ENGINEER ECOSYSTEM — 200 TOUGH-LEVEL MCQs

Advanced practice question bank | Single-correct answer format

Question count	200
Difficulty	Hard / Tough
Question formats	Numericals, conceptual applications, assertion–reason, statement-based and matching
Answer placement	Separate answer key with concise explanations at the end

Instructions

1. Each question has four options (a–d) and only one best answer.
2. Attempt all questions before consulting the answer key.
3. For numerical questions, use the stated assumptions and units.
4. The answer key and one-line explanations begin after Question 200.

Coverage

Ecosystem structure and function; GPP, NPP and NEP; energy flow and ecological efficiencies; ecological pyramids; food chains and food webs; decomposition; nutrient cycles; succession; population and community ecology; species interactions; biodiversity; ecosystem stability; aquatic and terrestrial productivity; eutrophication; biomagnification; ecosystem services and restoration.

SECTION I — QUESTIONS

Q1. A lake ecosystem has GPP of 2,400 kJ m■² day■¹ and producer respiration of 900 kJ m■² day■¹. Its NPP is:

(a) 1,500 kJ m■² day■¹

(b) 2,400 kJ m■² day■¹

(c) 900 kJ m■² day■¹

(d) 3,300 kJ m■² day■¹

Q2. At each trophic transfer, 10% of energy is passed on. If producers contain 80,000 kJ, energy available to secondary consumers is:

(a) 80 kJ

(b) 8,000 kJ

(c) 800 kJ

(d) 8000? No transfer loss

Q3. Producers store 50,000 kJ. Transfer efficiencies are 12% from producers to herbivores, 15% to secondary consumers, and 10% to tertiary consumers. Energy at tertiary consumers is:

(a) 1,080 kJ

(b) 900 kJ

(c) 90 kJ

(d) 750 kJ

Q4. If producers fix 10,000 kJ and the producer-to-herbivore transfer efficiency is 10%, how much energy reaches herbivores?

(a) 1,000 kJ

(b) 1,000 kJ

(c) 10,000 kJ

(d) 9,000 kJ

Q5. An ecosystem has GPP of 1300 g C m■² yr■¹ and producer respiration of 360 g C m■² yr■¹. Calculate NPP.

(a) 1300 g C m■² yr■¹

(b) 360 g C m■² yr■¹

(c) 940 g C m■² yr■¹

(d) 1660 g C m■² yr■¹

Q6. With a 10% transfer efficiency at each step, how much of 66,000 kJ stored by producers reaches the third consumer level (three transfers from producers)?

(a) 6,600.0 kJ

(b) 6.6 kJ

(c) 660.0 kJ

(d) 66.0 kJ

Q7. If NPP is 1340 g C m■² yr■¹ and GPP is 2240 g C m■² yr■¹, producer respiration equals:

(a) 900 g C m■² yr■¹

(b) 3580 g C m■² yr■¹

(c) 2240 g C m■² yr■¹

(d) 1340 g C m■² yr■¹

Q8. A 2.4 ha grassland has an annual NPP of 600 g C m■². Using 1 ha = 10,000 m², total annual carbon production is:

(a) 14,400 kg C

(b) 14,400 kg C

(c) 1,440 kg C

(d) 144,000 kg C

Q9. A consumer assimilates 1000 kJ from food and loses 250.0 kJ in egestion. Assimilation efficiency is:

(a) 25.0%

(b) 400.0%

(c) 25.0%

(d) 75.0%

Q10. Herbivores consume 40% of a plant community's NPP of 1020 kJ m■² yr■¹. Energy ingested by herbivores is:

(a) 2550 kJ m■² yr■¹

(b) 408 kJ m■² yr■¹

(c) 612 kJ m■² yr■¹

(d) 1020 kJ m■² yr■¹

Q11. An ecosystem has GPP = 1850 and autotrophic respiration = 405 g C m■² yr■¹. The NPP:GPP ratio is approximately:

(a) 0.78

(b) 0.22

(c) 1.28

(d) 1.28

Q12. Producers store 16,000 kJ. If transfer efficiency is 10% to herbivores and 20% to carnivores, energy at the carnivore level is:

(a) 3,200 kJ

(b) 320 kJ

(c) 3,200 kJ

(d) 1,600 kJ

Q13. A plant stand fixes 3.30 kg C m■² yr■¹ gross and retains 2.32 kg C m■² yr■¹ net. Its autotrophic respiration is:

(a) 2.32 kg C m■² yr■¹

(b) 3.30 kg C m■² yr■¹

(c) 5.62 kg C m■² yr■¹

(d) 0.98 kg C m■² yr■¹

Q14. If producers fix 20,000 kJ and the producer-to-herbivore transfer efficiency is 12%, how much energy reaches herbivores?

(a) 2,400 kJ

(b) 1,667 kJ

(c) 17,600 kJ

(d) 24,000 kJ

Q15. An ecosystem has GPP of 2300 g C m■² yr■¹ and producer respiration of 460 g C m■² yr■¹. Calculate NPP.

(a) 2760 g C m■² yr■¹

(b) 1840 g C m■² yr■¹

(c) 460 g C m■² yr■¹

(d) 2300 g C m■² yr■¹

Q16. With a 10% transfer efficiency at each step, how much of 76,000 kJ stored by producers reaches the third consumer level (three transfers from producers)?

(a) 7.6 kJ

(b) 76.0 kJ

(c) 7,600.0 kJ

(d) 760.0 kJ

Q17. If NPP is 1640 g C m■² yr■¹ and GPP is 3040 g C m■² yr■¹, producer respiration equals:

(a) 1640 g C m■² yr■¹

(b) 3040 g C m■² yr■¹

(c) 4680 g C m■² yr■¹

(d) 1400 g C m■² yr■¹

Q18. A 3.4 ha grassland has an annual NPP of 850 g C m■². Using 1 ha = 10,000 m², total annual carbon production is:

(a) 289,000 kg C

(b) 28,900 kg C

(c) 2,890 kg C

(d) 28,900 kg C

Q19. A consumer assimilates 1400 kJ from food and loses 350.0 kJ in egestion. Assimilation efficiency is:

(a) 25.0%

(b) 75.0%

(c) 400.0%

(d) 25.0%

Q20. Herbivores consume 40% of a plant community's NPP of 1220 kJ m■² yr■¹. Energy ingested by herbivores is:

(a) 1220 kJ m■² yr■¹

(b) 3050 kJ m■² yr■¹

(c) 488 kJ m■² yr■¹

(d) 732 kJ m■² yr■¹

Q21. An ecosystem has GPP = 2350 and autotrophic respiration = 555 g C m■² yr■¹. The NPP:GPP ratio is approximately:

(a) 1.31

(b) 0.24

(c) 1.31

(d) 0.76

Q22. Producers store 21,000 kJ. If transfer efficiency is 10% to herbivores and 20% to carnivores, energy at the carnivore level is:

(a) 2,100 kJ

(b) 4,200 kJ

(c) 4,200 kJ

(d) 420 kJ

Q23. A plant stand fixes 4.30 kg C m■² yr■¹ gross and retains 3.12 kg C m■² yr■¹ net. Its autotrophic respiration is:

(a) 1.18 kg C m■² yr■¹

(b) 3.12 kg C m■² yr■¹

(c) 4.30 kg C m■² yr■¹

(d) 7.42 kg C m■² yr■¹

Q24. If producers fix 30,000 kJ and the producer-to-herbivore transfer efficiency is 14%, how much energy reaches herbivores?

(a) 25,800 kJ

(b) 4,200 kJ

(c) 2,143 kJ

(d) 42,000 kJ

Q25. An ecosystem has GPP of 3300 g C m■² yr■¹ and producer respiration of 560 g C m■² yr■¹. Calculate NPP.

(a) 2740 g C m■² yr■¹

(b) 3860 g C m■² yr■¹

(c) 3300 g C m■² yr■¹

(d) 560 g C m■² yr■¹

Q26. With a 10% transfer efficiency at each step, how much of 86,000 kJ stored by producers reaches the third consumer level (three transfers from producers)?

(a) 860.0 kJ

(b) 8.6 kJ

(c) 86.0 kJ

(d) 8,600.0 kJ

Q27. If NPP is 1940 g C m■² yr■¹ and GPP is 3840 g C m■² yr■¹, producer respiration equals:

(a) 3840 g C m■² yr■¹

(b) 1900 g C m■² yr■¹

(c) 5780 g C m■² yr■¹

(d) 1940 g C m■² yr■¹

Q28. A 4.4 ha grassland has an annual NPP of 1100 g C m■². Using 1 ha = 10,000 m², total annual carbon production is:

(a) 48,400 kg C

(b) 4,840 kg C

(c) 484,000 kg C

(d) 48,400 kg C

Q29. A consumer assimilates 1800 kJ from food and loses 450.0 kJ in egestion. Assimilation efficiency is:

(a) 400.0%

(b) 75.0%

(c) 25.0%

(d) 25.0%

Q30. Herbivores consume 40% of a plant community's NPP of 1420 kJ m■² yr■¹. Energy ingested by herbivores is:

(a) 852 kJ m■² yr■¹

(b) 1420 kJ m■² yr■¹

(c) 568 kJ m■² yr■¹

(d) 3550 kJ m■² yr■¹

Q31. An ecosystem has GPP = 2850 and autotrophic respiration = 705 g C m■² yr■¹. The NPP:GPP ratio is approximately:

(a) 0.75

(b) 0.25

(c) 1.33

(d) 1.33

Q32. Producers store 26,000 kJ. If transfer efficiency is 10% to herbivores and 20% to carnivores, energy at the carnivore level is:

(a) 5,200 kJ

(b) 2,600 kJ

(c) 520 kJ

(d) 5,200 kJ

Q33. A plant stand fixes 5.30 kg C m■² yr■¹ gross and retains 3.92 kg C m■² yr■¹ net. Its autotrophic respiration is:

(a) 9.22 kg C m■² yr■¹

(b) 5.30 kg C m■² yr■¹

(c) 1.38 kg C m■² yr■¹

(d) 3.92 kg C m■² yr■¹

Q34. If producers fix 40,000 kJ and the producer-to-herbivore transfer efficiency is 10%, how much energy reaches herbivores?

(a) 40,000 kJ

(b) 4,000 kJ

(c) 36,000 kJ

(d) 4,000 kJ

Q35. An ecosystem has GPP of 4300 g C m■² yr■¹ and producer respiration of 660 g C m■² yr■¹. Calculate NPP.

(a) 4960 g C m■² yr■¹

(b) 660 g C m■² yr■¹

(c) 4300 g C m■² yr■¹

(d) 3640 g C m■² yr■¹

Q36. With a 10% transfer efficiency at each step, how much of 96,000 kJ stored by producers reaches the third consumer level (three transfers from producers)?

(a) 96.0 kJ

(b) 960.0 kJ

(c) 9.6 kJ

(d) 9,600.0 kJ

Q37. If NPP is 2240 g C m■² yr■¹ and GPP is 4640 g C m■² yr■¹, producer respiration equals:

(a) 6880 g C m■² yr■¹

(b) 2400 g C m■² yr■¹

(c) 2240 g C m■² yr■¹

(d) 4640 g C m■² yr■¹

Q38. A 5.4 ha grassland has an annual NPP of 1350 g C m■². Using 1 ha = 10,000 m², total annual carbon production is:

(a) 729,000 kg C

(b) 72,900 kg C

(c) 7,290 kg C

(d) 72,900 kg C

Q39. A consumer assimilates 2200 kJ from food and loses 550.0 kJ in egestion. Assimilation efficiency is:

(a) 75.0%

(b) 25.0%

(c) 400.0%

(d) 25.0%

Q40. Herbivores consume 40% of a plant community's NPP of 1620 kJ m■² yr■¹. Energy ingested by herbivores is:

(a) 1620 kJ m■² yr■¹

(b) 648 kJ m■² yr■¹

(c) 972 kJ m■² yr■¹

(d) 4050 kJ m■² yr■¹

Q41. An ecosystem has GPP = 3350 and autotrophic respiration = 855 g C m■² yr■¹. The NPP:GPP ratio is approximately:

(a) 0.26

(b) 1.34

(c) 0.74

(d) 1.34

Q42. Producers store 31,000 kJ. If transfer efficiency is 10% to herbivores and 20% to carnivores, energy at the carnivore level is:

(a) 6,200 kJ

(b) 3,100 kJ

(c) 6,200 kJ

(d) 620 kJ

Q43. A plant stand fixes 6.30 kg C m■² yr■¹ gross and retains 4.72 kg C m■² yr■¹ net. Its autotrophic respiration is:

(a) 6.30 kg C m■² yr■¹

(b) 11.02 kg C m■² yr■¹

(c) 1.58 kg C m■² yr■¹

(d) 4.72 kg C m■² yr■¹

Q44. If producers fix 50,000 kJ and the producer-to-herbivore transfer efficiency is 12%, how much energy reaches herbivores?

(a) 44,000 kJ

(b) 6,000 kJ

(c) 4,167 kJ

(d) 60,000 kJ

Q45. An ecosystem has GPP of 5300 g C m■² yr■¹ and producer respiration of 760 g C m■² yr■¹. Calculate NPP.

(a) 6060 g C m■² yr■¹

(b) 5300 g C m■² yr■¹

(c) 4540 g C m■² yr■¹

(d) 760 g C m■² yr■¹

Q46. With a 10% transfer efficiency at each step, how much of 106,000 kJ stored by producers reaches the third consumer level (three transfers from producers)?

(a) 10,600.0 kJ

(b) 10.6 kJ

(c) 1,060.0 kJ

(d) 106.0 kJ

Q47. If NPP is 2540 g C m■² yr■¹ and GPP is 5440 g C m■² yr■¹, producer respiration equals:

(a) 5440 g C m■² yr■¹

(b) 7980 g C m■² yr■¹

(c) 2540 g C m■² yr■¹

(d) 2900 g C m■² yr■¹

Q48. A 6.4 ha grassland has an annual NPP of 1600 g C m■². Using 1 ha = 10,000 m², total annual carbon production is:

(a) 10,240 kg C

(b) 1,024,000 kg C

(c) 102,400 kg C

(d) 102,400 kg C

Q49. A consumer assimilates 2600 kJ from food and loses 650.0 kJ in egestion. Assimilation efficiency is:

(a) 25.0%

(b) 400.0%

(c) 75.0%

(d) 25.0%

Q50. Herbivores consume 40% of a plant community's NPP of 1820 kJ m■² yr■¹. Energy ingested by herbivores is:

(a) 1092 kJ m■² yr■¹

(b) 4550 kJ m■² yr■¹

(c) 1820 kJ m■² yr■¹

(d) 728 kJ m■² yr■¹

Q51. Which statement best distinguishes standing crop from standing state?

(a) Both refer only to producer biomass

(b) Standing crop is biomass at a time; standing state is the amount of inorganic nutrients present at a time

(c) Standing state is energy stored in consumers

(d) Standing crop is annual productivity; standing state is annual nutrient uptake

Q52. Which statement about GPP and NPP is correct?

(a) NPP is GPP minus autotrophic respiration

(b) GPP is always lower than NPP

(c) NPP includes heterotrophic respiration

(d) NPP is the total solar energy intercepted

Q53. A food chain is unusually short primarily because:

(a) All consumers are ectothermic

(b) Energy is dissipated at each trophic transfer

(c) Producers are always less numerous than consumers

(d) Matter cannot be recycled

Q54. Which process most directly causes biomagnification?

(a) Accumulation of a pollutant within one organism over time

(b) Rapid biodegradation of a chemical in sediments

(c) Progressive increase in concentration of persistent, poorly metabolised substances at higher trophic levels

(d) Dilution of pollutants in water

Q55. Bioaccumulation differs from biomagnification because bioaccumulation:

(a) Requires a food chain with at least four trophic levels

(b) Always increases concentration at every higher trophic level

(c) Occurs only in producers

(d) Occurs within an organism over time from all exposure routes

Q56. Which ecological pyramid is necessarily upright in a stable ecosystem?

(a) All three

(b) Pyramid of biomass

(c) Energy pyramid

(d) Pyramid of numbers

Q57. An inverted biomass pyramid is most characteristic of:

(a) Deserts with sparse shrubs

(b) Aquatic systems dominated by rapidly turning-over phytoplankton

(c) Mature tropical forests

(d) Grasslands with large perennial grasses

Q58. Which statement about detritus food chains is most accurate?

(a) They always contain more trophic levels than grazing chains

(b) They operate only in aquatic ecosystems

(c) They begin with dead organic matter and are driven by decomposers and detritivores

(d) They do not contribute to nutrient cycling

Q59. The major ecological significance of a keystone species is that it:

(a) Has a disproportionately large effect relative to its abundance

(b) Is always the most abundant species

(c) Occupies the highest trophic level

(d) Has the greatest biomass in the community

Q60. Which is the best example of an ecosystem engineer?

(a) A beaver altering stream flow by building dams

(b) A hawk consuming rodents

(c) A fungus decomposing leaf litter

(d) A parasite feeding on a host

Q61. In ecological succession, facilitation means that:

(a) Late species prevent all early species from surviving

(b) Early species modify conditions making establishment of later species easier

(c) Disturbance permanently halts succession

(d) Species colonise independently of environmental change

Q62. Which statement correctly describes a sere?

(a) A sequence of communities during succession

(b) A group of organisms with identical niches

(c) The rate of nutrient cycling

(d) The final stable biomass of a trophic level

Q63. Primary succession differs from secondary succession mainly because primary succession:

(a) Begins after fire in a forest with soil intact

(b) Always proceeds faster

(c) Begins on a surface lacking developed soil

(d) Occurs only in aquatic habitats

Q64. Which is a likely pioneer group on newly exposed bare rock?

(a) Large herbivorous mammals

(b) Lichens and microbes

(c) Large canopy trees

(d) Deep-rooted climax shrubs

Q65. The term climax community is best treated as:

(a) A universally fixed endpoint independent of climate

(b) A community with zero species turnover

(c) A community that cannot be altered by disturbance

(d) A context-dependent relatively persistent stage under a given disturbance regime

Q66. Which statement about ecological niche is correct?

(a) Two species can never share any niche dimensions

(b) It is determined only by trophic position

(c) It includes the range of conditions and resources under which a species persists and reproduces

(d) It is synonymous with the physical place occupied

Q67. Competitive exclusion is most likely when two species:

(a) Occupy completely different niches

(b) Are separated geographically

(c) Use non-limiting resources

(d) Use the same limiting resource in the same way and one has a sustained competitive advantage

Q68. Resource partitioning promotes coexistence by:

(a) Making both species occupy identical microhabitats

(b) Increasing overlap in limiting-resource use

(c) Reducing direct competition through differentiated resource use

(d) Eliminating predation

Q69. Which interaction is correctly classified as mutualism?

(a) Both interacting species benefit

(b) One benefits and the other is unaffected

(c) One benefits and the other is harmed

(d) Both are harmed

Q70. Commensalism is represented by:

(a) +/–

(b) +/+

(c) –/–

(d) +/0

Q71. Amensalism is represented by:

(a) –/0

(b) +/0

(c) –/–

(d) +/+

Q72. Which statement about parasitism is most accurate?

(a) The host always dies immediately

(b) Both species benefit

(c) The parasite benefits while the host is harmed, usually without immediate host death being necessary

(d) It is identical to predation in all ecological respects

Q73. A Type III survivorship curve is characterised by:

(a) High mortality early in life followed by greater survival of those reaching adulthood

(b) Constant mortality at all ages

(c) Low juvenile mortality and high old-age mortality only

(d) No mortality until senescence

Q74. A Type II survivorship curve indicates:

(a) A population with no age structure

(b) Very high juvenile mortality only

(c) Mortality only in old age

(d) Approximately constant mortality risk across age classes

Q75. Which population parameter directly increases population size?

(a) Emigration

(b) Mortality

(c) Harvest

(d) Immigration

Q76. For a closed population over a time interval, the change in population size is:

(a) Immigration minus emigration

(b) Births minus deaths

(c) Deaths minus births

(d) Births plus deaths

Q77. The intrinsic rate of natural increase r is highest when:

(a) Death rate exceeds birth rate

(b) Per-capita birth rate exceeds per-capita death rate by the greatest amount

(c) Carrying capacity is zero

(d) Birth and death rates are equal

Q78. In logistic growth, population growth rate is maximal when population size is approximately:

(a) 2K

(b) Zero

(c) K

(d) K/2

Q79. At carrying capacity K in the logistic model:

(a) Net population growth is zero

(b) All individuals die

(c) Birth rate must be zero

(d) The population has no competition

Q80. Which statement about density-dependent regulation is correct?

(a) It is independent of density by definition

(b) Its per-capita effect generally changes with population density

(c) It always causes population extinction

(d) It includes only volcanic eruptions

Q81. A density-independent factor is best illustrated by:

(a) Territorial exclusion

(b) Competition for nesting sites

(c) Density-related disease transmission

(d) A severe frost affecting a population regardless of its density

Q82. The realized niche is often narrower than the fundamental niche because of:

(a) The species' trophic level alone

(b) Lack of abiotic conditions

(c) Biotic interactions such as competition and predation

(d) Absence of physiological tolerance limits

Q83. Which statement about ecosystem boundaries is most defensible?

(a) They are often defined operationally and may be relatively open to energy and matter flows

(b) Only natural systems qualify as ecosystems

(c) Every ecosystem is completely closed to matter

(d) Boundaries are always identical to political borders

Q84. Which process converts atmospheric nitrogen into biologically usable reduced nitrogen?

(a) Nitrification

(b) Denitrification

(c) Ammonification

(d) Biological nitrogen fixation

Q85. Nitrification is the:

(a) Reduction of nitrate to nitrogen gas

(b) Oxidation of ammonium to nitrite and then nitrate

(c) Conversion of organic nitrogen to ammonium

(d) Fixation of atmospheric nitrogen

Q86. Denitrification generally:

(a) Fixes N■ into organic nitrogen

(b) Oxidises ammonium to nitrate

(c) Reduces nitrate/nitrite to gaseous nitrogen forms under low-oxygen conditions

(d) Releases phosphate from apatite

Q87. Ammonification refers to:

(a) Conversion of ammonium to nitrate

(b) Conversion of nitrate to N■

(c) Conversion of organic nitrogen into ammonium

(d) Conversion of N■ to ammonia by lightning only

Q88. Which biogeochemical cycle lacks a major gaseous atmospheric reservoir?

(a) Oxygen cycle

(b) Nitrogen cycle

(c) Phosphorus cycle

(d) Carbon cycle

Q89. Eutrophication in freshwater is most commonly accelerated by excess:

(a) Sodium chloride loading

(b) Phosphorus loading

(c) Silica depletion alone

(d) Dissolved oxygen loading

Q90. A major consequence of decomposition-driven hypoxia after an algal bloom is:

(a) Immediate removal of all nutrients

(b) High microbial oxygen demand during breakdown of organic matter

(c) Suppression of all microbial activity

(d) Permanent increase in dissolved oxygen

Q91. Which statement about carbon sequestration is correct?

(a) Respiration has no influence on ecosystem carbon stocks

(b) Long-term storage depends on the balance of carbon inputs, outputs and residence time

(c) Gross photosynthesis alone determines long-term storage

(d) All carbon fixed by plants remains stored permanently

Q92. A system has NPP of 1,000 g C m■² yr■¹. If heterotrophic respiration is 700 g C m■² yr■¹, net ecosystem production (NEP) is:

(a) 300 g C m■² yr■¹

(b) −300 g C m■² yr■¹

(c) 1,700 g C m■² yr■¹

(d) 700 g C m■² yr■¹

Q93. Which factor most directly controls decomposition rate in warm, moist, oxygenated soils?

(a) Latitude alone

(b) Standing crop of top predators

(c) Temperature, moisture, substrate quality and decomposer activity

(d) The number of trophic levels

Q94. Lignin-rich litter generally decomposes more slowly because:

(a) It dissolves instantly in water

(b) It is always nitrogen-rich

(c) Its complex, resistant structure is difficult for many decomposers to break down

(d) It contains no carbon

Q95. Which ecosystem generally has high net primary productivity per unit area?

(a) Open ocean

(b) Hot desert

(c) Tropical wet forest

(d) Arctic tundra

Q96. Which statement best distinguishes productivity from standing biomass?

(a) Standing biomass is a rate per year

(b) Productivity is a rate of biomass/energy generation; standing biomass is a stock at a given time

(c) Both are stocks measured at one instant

(d) Productivity is always measured in kilograms only

Q97. Ecological efficiency can refer to:

(a) The proportion of energy transferred between trophic levels

(b) The ratio of rainfall to runoff

(c) The fraction of species that are endemic

(d) The rate of soil erosion only

Q98. Which statement about the 10% law is scientifically most appropriate?

(a) It is a useful approximate rule, not a universal fixed efficiency

(b) It means 10% of organisms survive

(c) It applies only to nutrient transfer

(d) Exactly 10% is transferred in every ecosystem and every link

Q99. Why can a grazing food chain be less conspicuous than a detrital pathway in some forests?

(a) Forests lack herbivores

(b) Energy is recycled from predators to plants

(c) A large fraction of plant production enters litter and dead organic matter

(d) Decomposers do not use plant material

Q100. Which statement about ecological resilience is correct?

(a) It is identical to resistance in every definition

(b) It concerns the capacity to absorb disturbance and retain essential structure or function

(c) It means a system never changes

(d) It is measured only by species richness

Q101. Resistance differs from resilience because resistance refers more to:

(a) The number of trophic levels

(b) The degree of change during disturbance

(c) The rate of primary production

(d) The speed of recovery after disturbance

Q102. Consider the statements: 1. Energy flow is unidirectional. 2. Nutrients may be recycled. 3. Energy is recycled in the same way as phosphorus. Which are correct?

(a) 1, 2 and 3

(b) 1 and 2 only

(c) 2 and 3 only

(d) 1 and 3 only

Q103. Assertion (A): The pyramid of energy is always upright. Reason (R): Energy is lost as heat and through metabolic processes at each trophic transfer.

(a) A is true, R is false

(b) Both A and R are true, but R does not explain A

(c) A is false, R is true

(d) Both A and R are true, and R correctly explains A

Q104. Assertion (A): A biomass pyramid in an aquatic ecosystem can be inverted. Reason (R): Phytoplankton can have high turnover and low standing biomass despite high production.

(a) Both A and R are true, but R does not explain A

(b) Both A and R are true, and R correctly explains A

(c) A is false, R is true

(d) A is true, R is false

Q105. Which pair is incorrectly matched?

(a) Denitrification — nitrate to gaseous N

(b) Nitrification — reduction of nitrate to N■

(c) Ammonification — organic N to ammonium

(d) Nitrogen fixation — N■ to ammonia

Q106. Match the term with its description: A. GPP; B. NPP; C. NEP. 1. GPP−Ra; 2. NPP−Rh; 3. Total photosynthetic carbon fixation.

(a) A-2, B-3, C-1

(b) A-1, B-2, C-3

(c) A-3, B-1, C-2

(d) A-3, B-2, C-1

Q107. Which of the following can cause an inverted pyramid of numbers?

(a) A desert with few shrubs and fewer herbivores

(b) One large tree supporting many herbivorous insects and parasites

(c) A pond with abundant phytoplankton

(d) A grassland with many producers and fewer herbivores

Q108. Which combination best describes a mature ecosystem's nutrient cycling?

(a) Complete absence of nutrient loss

(b) No decomposition

(c) Nutrients move only from soil to atmosphere

(d) Relatively tight internal cycling with losses and external inputs still occurring

Q109. Which is the correct order of processes in a simplified nitrogen cycle pathway from organic matter to gaseous N■?

(a) Denitrification → ammonification → fixation

(b) Fixation → denitrification → nitrification

(c) Nitrification → fixation → ammonification

(d) Ammonification → nitrification → denitrification

Q110. Which statement about mycorrhizae is most accurate?

(a) They fix atmospheric nitrogen in all plant species

(b) They commonly enhance plant phosphorus and water acquisition in exchange for plant carbon

(c) They are exclusively pathogenic

(d) They eliminate plant respiration

Q111. The ecological term 'ecotone' denotes:

(a) The final stage of succession

(b) A transition zone between adjacent ecological communities

(c) A nutrient reservoir in bedrock

(d) A trophic level between herbivores and carnivores

Q112. An edge effect refers to:

(a) The reduction of all diversity at every boundary

(b) Energy transfer from decomposers to producers

(c) Changes in ecological conditions and species composition near habitat boundaries

(d) A form of nitrogen fixation

Q113. Which statement about species richness and evenness is correct?

(a) Richness measures biomass; evenness measures productivity

(b) Evenness is the number of trophic levels

(c) Richness counts species; evenness describes relative abundance distribution

(d) Both are identical measures

Q114. A community has 4 species with proportions 0.7, 0.1, 0.1 and 0.1. Compared with a community with equal proportions, it has:

(a) Higher evenness

(b) Lower evenness

(c) Identical evenness

(d) Zero richness

Q115. Which statement about the Shannon diversity index H′ is correct?

(a) It measures only primary productivity

(b) It is independent of relative abundance

(c) It generally increases with both richness and evenness

(d) It can be calculated without species proportions

Q116. The intermediate disturbance hypothesis proposes that diversity may peak at:

(a) Intermediate disturbance frequency or intensity

(b) Only after complete ecosystem destruction

(c) The maximum possible disturbance

(d) Zero disturbance in all ecosystems

Q117. Which is an example of bottom-up control?

(a) Disease regulates a predator population

(b) Nutrient supply increases phytoplankton and propagates upward through consumers

(c) A top predator controls prey density

(d) Predator removal increases herbivore abundance and reduces plants

Q118. Top-down trophic cascade is best represented by:

(a) Predator increase → herbivore decrease → plant biomass increase

(b) Nutrient increase → producer increase → herbivore increase

(c) Decomposer increase → litter decrease only

(d) Rainfall increase → soil moisture increase

Q119. Which statement about biomagnification is most likely true for methylmercury?

(a) It accumulates only in plant roots

(b) It cannot enter aquatic food webs

(c) Concentrations can increase in organisms at higher trophic positions

(d) It is rapidly metabolised and excreted by all fish

Q120. A pollutant concentration rises in an individual fish as it ages, but does not increase consistently across trophic levels. This is evidence primarily of:

(a) Biodegradation

(b) Bioaccumulation

(c) Biostimulation

(d) Biomagnification

Q121. Which factor makes a chemical more likely to biomagnify?

(a) Immediate microbial mineralisation

(b) High volatility alone

(c) Rapid degradation and high water solubility with rapid excretion

(d) Persistence, bioavailability and affinity for lipid-rich tissues

Q122. Which is the most defensible statement about ecosystem services?

(a) They exclude climate regulation

(b) They include provisioning, regulating, cultural and supporting or habitat-related functions

(c) They are independent of biodiversity

(d) They refer only to marketable timber and crops

Q123. Wetland removal of nitrate from water is often promoted by:

(a) Photosynthesis by fish

(b) Nitrification alone under fully oxic conditions

(c) Denitrification in anoxic sediments

(d) Evaporation of nitrate

Q124. Which statement about phosphorus limitation is most accurate?

(a) It is common in many freshwater systems, but limitation varies with context

(b) It is caused by atmospheric N■

(c) It can never be altered by human inputs

(d) It occurs in every ecosystem at all times

Q125. In a lake, thermal stratification can promote bottom-water hypoxia when:

(a) Water is completely mixed continuously

(b) Nutrients are absent

(c) Surface photosynthesis supplies oxygen directly to deep water at all times

(d) Organic matter decomposition consumes oxygen and stratification limits replenishment

Q126. Which statement about ecological pyramids is correct?

(a) Energy pyramids can invert in forests

(b) All ecological pyramids must be upright

(c) Pyramids of numbers and biomass may be inverted; energy pyramids cannot

(d) Biomass pyramids always show energy flow

Q127. A food web is generally more realistic than a simple food chain because it:

(a) Assumes each species has one food source

(b) Excludes omnivory

(c) Shows only one linear transfer route

(d) Represents multiple feeding links and alternative energy pathways

Q128. Omnivory complicates trophic-level assignment because an omnivore:

(a) Is always a decomposer

(b) Cannot transfer energy

(c) Feeds across more than one trophic level

(d) Consumes only producers

Q129. Which organism is most directly a detritivore rather than a decomposer in the strict functional sense?

(a) Mould decomposing cellulose externally

(b) Saprotrophic fungus secreting extracellular enzymes

(c) Earthworm fragmenting and ingesting litter

(d) Bacterium mineralising dissolved organic matter

Q130. Which statement about ecological efficiency is correct?

(a) Assimilation efficiency, production efficiency and consumption efficiency describe different steps

(b) They are interchangeable terms for 10% transfer

(c) Production efficiency includes unconsumed food

(d) Consumption efficiency is the fraction of assimilated energy converted to biomass

Q131. Which sequence correctly represents energy partitions for an animal consumer?

(a) Ingested = production + respiration only

(b) Ingested energy = assimilated + egested; assimilated = production + respiration

(c) Assimilated = ingested + egested

(d) Production = respiration − assimilation

Q132. Which statement about ecological stoichiometry is correct?

(a) It concerns only the number of species in a habitat

(b) It is another name for trophic pyramids

(c) It excludes nutrient limitation

(d) It examines how elemental ratios in organisms and resources influence ecological interactions and processes

Q133. A high C:N ratio in plant litter generally tends to:

(a) Accelerate nitrogen mineralisation in all conditions

(b) Convert phosphorus directly to nitrate

(c) Eliminate microbial demand for nitrogen

(d) Slow decomposition and promote microbial nitrogen immobilisation

Q134. Which statement about biological control is correct?

(a) Introducing a natural enemy can suppress a pest but may create non-target ecological risks

(b) It always eliminates the pest permanently

(c) It is equivalent to pesticide bioaccumulation

(d) It has no ecological interactions

Q135. Which ecosystem is most likely to have low annual NPP per unit area?

(a) Hot desert

(b) Tropical rainforest

(c) Estuary

(d) Tropical swamp forest

Q136. Why can estuaries have high productivity?

(a) They are always oxygen-saturated at all depths

(b) Salinity prevents all photosynthesis

(c) Nutrient inputs, shallow light penetration and mixing can support high primary production

(d) They have no nutrient input

Q137. Which statement about ocean productivity is correct?

(a) Open ocean has the highest productivity per unit area everywhere

(b) Ocean productivity is independent of nutrients and light

(c) Phytoplankton are absent from oligotrophic waters

(d) Open-ocean productivity per unit area is often low, but total global production is large due to ocean area

Q138. Upwelling often increases marine productivity because it:

(a) Raises salinity until phytoplankton stop growing

(b) Removes all dissolved nutrients

(c) Brings nutrient-rich deep water into the sunlit surface layer

(d) Prevents mixing

Q139. Which statement about the euphotic zone is correct?

(a) It contains no phytoplankton

(b) It is the water layer receiving sufficient light for net photosynthesis

(c) It is always identical to the thermocline

(d) It is defined by complete darkness

Q140. In a forest ecosystem, a large fraction of net primary production may be allocated to:

(a) Woody biomass and roots as well as leaves and reproductive tissues

(b) Only atmospheric oxygen

(c) Only animal biomass

(d) Only edible leaves consumed immediately by herbivores

Q141. Which is the strongest reason that gross primary productivity cannot be inferred from standing biomass alone?

(a) GPP excludes photosynthesis

(b) Standing biomass is a stock, whereas GPP is a flux and turnover rates vary

(c) Biomass is measured only in energy units

(d) Standing biomass always equals annual GPP

Q142. A system with positive NEP is, over the measurement interval, generally:

(a) A net carbon sink for the ecosystem-atmosphere exchange, absent other unaccounted fluxes

(b) At exact carbon equilibrium

(c) Unable to respire

(d) A net carbon source by definition

Q143. Which is an example of a positive feedback in ecosystem change?

(a) Higher plant cover reduces erosion

(b) Predators suppress herbivores

(c) Warming increases permafrost thaw, releasing greenhouse gases that promote further warming

(d) Increased shade reduces soil temperature

Q144. Which statement about alternative stable states is correct?

(a) They occur only in laboratory microcosms

(b) A system may persist in different configurations under similar external conditions, with transitions triggered by disturbance or feedbacks

(c) They mean that succession does not occur

(d) Every ecosystem has exactly one inevitable stable state

Q145. Which process is most likely to increase soil organic carbon over the long term?

(a) Increased erosion of topsoil

(b) Inputs of plant residues exceeding decomposition and export losses

(c) Higher decomposition than input

(d) Complete removal of litter

Q146. Which statement about ecological restoration is most accurate?

(a) Reference conditions guide goals, but restored systems need not reproduce an exact historical state

(b) Species introductions are always beneficial

(c) Restoration always returns an ecosystem to a pristine state

(d) Hydrology is irrelevant to wetland restoration

Q147. Which factor is especially important in determining whether a disturbed ecosystem undergoes secondary succession?

(a) Survival of soil, seed bank, roots and nearby propagule sources

(b) Complete absence of soil

(c) Formation of new bedrock

(d) Elimination of all organic matter

Q148. Which statement about invasive species is correct?

(a) All non-native species inevitably cause ecological harm

(b) Native species cannot become invasive

(c) Not every introduced species becomes invasive; impact depends on establishment, spread and ecological effects

(d) Invasive species are always predators

Q149. A species with a broad tolerance range for environmental conditions is described as:

(a) Obligate parasite

(b) Stenotopic

(c) Eurytopic

(d) Endemic

Q150. Which statement about Liebig's law of the minimum is correct?

(a) Growth is constrained by the essential resource in shortest supply relative to demand

(b) Only sunlight can be limiting

(c) The most abundant nutrient always limits growth

(d) All resources contribute equally regardless of availability

Q151. Shelford's law of tolerance emphasises that:

(a) Species have unlimited tolerance to temperature

(b) Biotic interactions are irrelevant

(c) Both deficiency and excess of an environmental factor can limit species performance

(d) Only low levels of a factor are harmful

Q152. In a simple ecosystem carbon budget, GPP = 2,800, autotrophic respiration = 1,100 and heterotrophic respiration = 1,300 g C m■² yr■¹. NEP equals:

(a) −400 g C m■² yr■¹

(b) 400 g C m■² yr■¹

(c) 1,500 g C m■² yr■¹

(d) 1,700 g C m■² yr■¹

Q153. A herbivore ingests 1,000 kJ, egests 300 kJ and respires 500 kJ. Its secondary production is:

(a) 700 kJ

(b) 800 kJ

(c) 500 kJ

(d) 200 kJ

Q154. A population grows from 200 to 242 individuals in one year in the absence of migration. If exponential growth is assumed, the approximate discrete per-capita growth rate is:

(a) 0.10 yr■¹

(b) 0.21 yr■¹

(c) 0.17 yr■¹

(d) 0.42 yr■¹

Q155. A logistic population has r = 0.4 yr■¹, K = 1,000 and N = 500. Its instantaneous growth rate dN/dt is:

(a) 200 individuals yr■¹

(b) 400 individuals yr■¹

(c) 50 individuals yr■¹

(d) 100 individuals yr■¹

Q156. A producer trophic level stores 20,000 kJ. If 15% reaches herbivores and 8% of herbivore energy reaches carnivores, carnivore energy is:

(a) 300 kJ

(b) 240 kJ

(c) 1,600 kJ

(d) 24 kJ

Q157. A plant community's GPP is 3,600 g C m■² yr■¹ and NPP is 2,160. Autotrophic respiration as a fraction of GPP is:

(a) 1.67%

(b) 25%

(c) 40%

(d) 60%

Q158. If a chemical has a bioconcentration factor of 500 L kg■¹ and water concentration is 0.004 mg L■¹, predicted organism concentration at equilibrium from water exposure alone is:

(a) 0.002 mg kg■¹

(b) 0.5 mg kg■¹

(c) 2 mg kg■¹

(d) 125 mg kg■¹

Q159. A community has 10 species. A diversity measure that is sensitive to dominance by the most abundant species will generally be lower when:

(a) All species have equal abundance

(b) Rare species become more evenly represented

(c) The abundance distribution becomes more even

(d) One species accounts for 90% of individuals

Q160. Which statement about the biological pump in oceans is correct?

(a) Sinking organic matter transfers carbon from surface waters to the deep ocean

(b) It prevents all ocean-atmosphere CO■ exchange

(c) It is driven exclusively by fish migration

(d) It transfers only heat and no carbon

Q161. Which of the following best explains why energy pyramids use energy per unit area per unit time?

(a) It removes the need to identify trophic levels

(b) It represents the rate of energy transfer or production, not merely a standing stock

(c) It measures only organism counts

(d) It is equivalent to biomass at one instant

Q162. A forest is clear-felled but the soil profile and seed bank remain. The ensuing recovery is best classified as:

(a) Secondary succession

(b) No succession

(c) Hydrarch succession

(d) Primary succession

Q163. A newly formed volcanic island has bare rock and no soil. The likely sequence begins with:

(a) A climax community before any colonisation

(b) A detritus food chain with deep humus

(c) Weathering and pioneer colonisation followed by soil development

(d) Mature forest establishment immediately

Q164. Which statement about decomposition is incorrect?

(a) Decomposition is independent of oxygen availability

(b) Fragmentation increases surface area

(c) Leaching removes soluble compounds

(d) Catabolism releases inorganic nutrients

Q165. In a stratified lake, the hypolimnion becomes anoxic. Which process can continue and contribute to nutrient release?

(a) Atmospheric gas exchange directly with deep water

(b) High-light photosynthesis

(c) Anaerobic microbial decomposition and redox-sensitive sediment reactions

(d) Aerobic nitrification at high rates throughout

Q166. Which is the most likely outcome of adding a large phosphorus load to a P-limited lake?

(a) Nitrogen fixation stops instantly

(b) The lake becomes oligotrophic

(c) Primary production necessarily falls

(d) Algal production may rise, followed by increased decomposition and oxygen depletion

Q167. Which statement about trophic cascades is most precise?

(a) They are synonymous with nutrient cycling

(b) Effects of consumers can propagate across multiple trophic levels, sometimes indirectly

(c) They require exactly four trophic levels

(d) They occur only in terrestrial ecosystems

Q168. A parasite that changes host behaviour and increases predation risk can influence:

(a) Only abiotic nutrient pools

(b) Energy flow and interaction structure beyond the host-parasite pair

(c) No other species

(d) Only host body mass, never food webs

Q169. Which statement about a functional response is correct?

(a) It describes how an individual's consumption rate changes with prey density

(b) It describes population size against carrying capacity

(c) It measures decomposition of litter

(d) It is identical to a survivorship curve

Q170. A Type II predator functional response is characterised by:

(a) Consumption rate declining linearly with prey density

(b) Consumption rate increasing without limit

(c) Consumption rate rising with prey density but eventually saturating

(d) A fixed consumption rate at all prey densities

Q171. Which statement about ecological succession after repeated severe disturbance is correct?

(a) Disturbance removes environmental filtering

(b) Disturbance always increases biomass

(c) Succession must always reach the same climax forest

(d) The system may remain in an early-successional state if disturbance prevents later stages

Q172. Which process is a major pathway of phosphorus loss from terrestrial ecosystems?

(a) Methane oxidation

(b) Volatilisation of phosphate gas

(c) Denitrification

(d) Erosion and runoff of phosphate-bearing soil and particles

Q173. Which is the best example of nutrient immobilisation?

(a) Microbes release ammonium from proteins

(b) Microbes incorporate inorganic ammonium into their biomass while decomposing carbon-rich litter

(c) Plants release oxygen during photosynthesis

(d) Nitrate is reduced to N■

Q174. Which statement about carbon-to-nitrogen ratio in litter is generally correct?

(a) C:N has no effect on decomposition

(b) Low C:N always prevents decomposition

(c) High C:N often indicates lower litter nitrogen availability to decomposers

(d) C:N is a measure of trophic transfer efficiency

Q175. An ecosystem with high productivity but low standing biomass is possible when:

(a) Production is zero

(b) Biomass turnover is rapid

(c) All biomass is permanently stored

(d) Consumers are absent

Q176. Which condition favours high primary production in aquatic ecosystems?

(a) High turbidity and severe nutrient depletion

(b) No dissolved inorganic carbon

(c) Adequate light and nutrient availability in the photic zone

(d) Permanent darkness

Q177. Why is silica important in many marine food webs?

(a) It is the main energy source for zooplankton

(b) It is required by all fish scales

(c) Diatoms require it to build siliceous frustules

(d) It replaces nitrogen fixation

Q178. Which statement about ecological footprint is correct?

(a) It is the same as a trophic pyramid

(b) It estimates biologically productive area needed to support resource use and absorb wastes under specified assumptions

(c) It measures only the number of species

(d) It directly measures ecosystem resilience

Q179. Which process can reduce dissolved oxygen in a receiving water body after wastewater discharge?

(a) Nitrate fixation by plants

(b) Evaporation of dissolved oxygen

(c) Microbial oxidation of biodegradable organic matter

(d) Photolysis of water

Q180. A high BOD in a stream most directly indicates:

(a) High primary productivity necessarily

(b) High potential oxygen demand from biodegradable organic matter

(c) Absence of microbes

(d) High dissolved oxygen concentration

Q181. Which statement about ecological indicators is correct?

(a) A bioindicator's response can integrate exposure over time, but interpretation depends on sensitivity and confounders

(b) Bioindicators remove pollutants from ecosystems

(c) Indicator species are always keystone species

(d) Every bioindicator responds only to one pollutant

Q182. A species restricted to a small geographic area is termed:

(a) Endemic

(b) Cosmopolitan

(c) Invasive by definition

(d) Eurytopic

Q183. Which statement about island biogeography is correct?

(a) All islands have equal richness

(b) Isolation increases immigration

(c) Species richness is influenced by island area and isolation through immigration and extinction rates

(d) Area has no influence on extinction

Q184. In the species-area relationship S = cA■, the exponent z describes:

(a) The number of trophic levels

(b) The slope of the log species–log area relationship

(c) The proportion of endemic species

(d) The carrying capacity

Q185. A metapopulation consists of:

(a) Only species with no extinction

(b) A community without immigration

(c) Spatially separated local populations connected by dispersal

(d) One completely mixed population

Q186. Which condition can promote coexistence through the storage effect?

(a) Complete absence of environmental variation

(b) Permanent competitive superiority of one species

(c) Identical responses to all environmental conditions

(d) Temporal environmental variation combined with species-specific responses and a persistent life stage or reservoir

Q187. Which statement about ecosystem stability is correct?

(a) Stability and productivity are identical

(b) It is adequately described by species richness alone

(c) A stable ecosystem has no fluctuations

(d) Stability has multiple dimensions, including resistance, resilience and persistence

Q188. Which statement about a disturbance regime is correct?

(a) Disturbance always reduces biodiversity

(b) Only disturbance intensity matters

(c) Frequency, intensity, duration, size and timing shape ecological responses

(d) Disturbance is always anthropogenic

Q189. Which ecosystem is especially adapted to periodic fire as a natural disturbance?

(a) All polar deserts

(b) All tropical rainforests

(c) Many savannas and fire-prone grasslands

(d) All deep lakes

Q190. Which statement about ecological succession and soil development is correct?

(a) Soil development can alter water and nutrient availability, changing later colonisation opportunities

(b) Plants do not influence soil formation

(c) Only animals drive soil development

(d) Soil properties remain fixed throughout succession

Q191. Which is a key distinction between habitat and niche?

(a) Niche refers only to shelter

(b) Habitat is where an organism lives; niche describes how it lives and functions in the ecosystem

(c) Habitat is a species' trophic level; niche is its geographic range only

(d) They are interchangeable

Q192. Which statement about density-dependent mortality is correct?

(a) It cannot regulate population size

(b) It may increase with crowding through competition, disease or predation

(c) It is necessarily constant across densities

(d) It is caused only by storms

Q193. A population has b = 0.32 and d = 0.47 per individual per year. Its intrinsic rate r is:

(a) 0.15 yr■¹

(b) –0.15 yr■¹

(c) 0.79 yr■¹

(d) –0.79 yr■¹

Q194. Which statement about carrying capacity is most accurate?

(a) It is independent of human activity

(b) It can change over time as resources, habitat and interactions change

(c) It is always equal to current population size

(d) It is a fixed species constant

Q195. Which statement about r-selected and K-selected strategies is most defensible?

(a) They are heuristic ends of a continuum, not rigid categories for all species

(b) Every species fits exclusively into one of two immutable groups

(c) K-selected species always have zero juvenile mortality

(d) r-selected species never compete

Q196. Which of the following is most likely to be a bottom-up effect of nitrogen deposition in a nitrogen-limited grassland?

(a) Increased plant production and possible shifts in competitive dominance

(b) A direct increase in top-predator biomass without food-web mediation

(c) Immediate elimination of decomposers

(d) No change in nutrient cycling

Q197. Which statement about nitrogen deposition is correct?

(a) It can cause acidification, eutrophication and biodiversity shifts in sensitive ecosystems

(b) It has no effect on soil chemistry

(c) It always improves biodiversity

(d) It adds phosphorus rather than nitrogen

Q198. Which pathway returns carbon from organisms to the atmosphere most directly?

(a) Nitrification

(b) Cellular respiration and decomposition

(c) Phosphorus weathering

(d) Nitrogen fixation

Q199. Which statement about methane in wetlands is correct?

(a) Methane oxidation produces nitrogen gas

(b) Wetlands cannot emit greenhouse gases

(c) Methane is produced only by plants in oxygenated leaves

(d) Anaerobic decomposition can produce methane, while oxidation near oxic interfaces can consume some methane

Q200. Which statement about the oxygen cycle is correct?

(a) Only animals produce oxygen

(b) Oxygen has no link to carbon cycling

(c) Oxygen is never exchanged between atmosphere and water

(d) Photosynthesis is a major source of molecular oxygen, while respiration and oxidation consume it

SECTION II — ANSWER KEY WITH EXPLANATIONS

Correct option and concise reasoning for each question.

1. a — NPP = GPP − autotrophic respiration = 2,400 − 900 = 1,500.

2. c — Producer → primary consumer = 8,000; primary → secondary = 800 kJ.

3. c — 50,000 × 0.12 × 0.15 × 0.10 = 90 kJ.

4. a — Energy transferred = 10,000 × 10/100 = 1,000 kJ.

5. c — NPP = GPP − R = 1300 − 360 = 940 g C m■² yr■¹.

6. d — Three transfers: 66,000 × 0.1^3 = 66.0 kJ.

7. a — R = GPP − NPP = 2240 − 1340 = 900.

8. a — 2.4 ha × 10,000 m²/ha × 600 g/m² = 14,400 kg C.

9. d — Assimilation efficiency = (assimilated − egested)/ingested × 100 = (1000−250.0)/1000 ×100 = 75.0%.

10. b — Consumed fraction = 0.40 × 1020 = 408 kJ m■² yr■¹.

11. a — NPP = 1850−405=1445; NPP:GPP = 1445/1850 = 0.78.

12. b — 16,000 × 0.10 × 0.20 = 320 kJ.

13. d — Respiration = GPP − NPP = 3.30 − 2.32 = 0.98.

14. a — Energy transferred = 20,000 × 12/100 = 2,400 kJ.

15. b — NPP = GPP − R = 2300 − 460 = 1840 g C m■² yr■¹.

16. b — Three transfers: 76,000 × 0.1^3 = 76.0 kJ.

17. d — R = GPP − NPP = 3040 − 1640 = 1400.

18. b — 3.4 ha × 10,000 m²/ha × 850 g/m² = 28,900 kg C.

19. b — Assimilation efficiency = (assimilated − egested)/ingested × 100 = (1400−350.0)/1400 ×100 = 75.0%.

20. c — Consumed fraction = 0.40 × 1220 = 488 kJ m■² yr■¹.

21. d — NPP = 2350−555=1795; NPP:GPP = 1795/2350 = 0.76.

22. d — 21,000 × 0.10 × 0.20 = 420 kJ.

23. a — Respiration = GPP − NPP = 4.30 − 3.12 = 1.18.

24. b — Energy transferred = 30,000 × 14/100 = 4,200 kJ.

25. a — NPP = GPP − R = 3300 − 560 = 2740 g C m■² yr■¹.

26. c — Three transfers: 86,000 × 0.1^3 = 86.0 kJ.

27. b — R = GPP − NPP = 3840 − 1940 = 1900.

28. a — 4.4 ha × 10,000 m²/ha × 1100 g/m² = 48,400 kg C.

29. b — Assimilation efficiency = (assimilated − egested)/ingested × 100 = (1800−450.0)/1800 ×100 = 75.0%.

30. c — Consumed fraction = 0.40 × 1420 = 568 kJ m■² yr■¹.

31. a — NPP = 2850−705=2145; NPP:GPP = 2145/2850 = 0.75.

32. c — 26,000 × 0.10 × 0.20 = 520 kJ.

33. c — Respiration = GPP − NPP = 5.30 − 3.92 = 1.38.

34. b — Energy transferred = 40,000 × 10/100 = 4,000 kJ.

35. d — NPP = GPP − R = 4300 − 660 = 3640 g C m■² yr■¹.

36. a — Three transfers: 96,000 × 0.1^3 = 96.0 kJ.

37. b — R = GPP − NPP = 4640 − 2240 = 2400.

38. b — 5.4 ha × 10,000 m²/ha × 1350 g/m² = 72,900 kg C.

39. a — Assimilation efficiency = (assimilated − egested)/ingested × 100 = (2200−550.0)/2200 ×100 = 75.0%.

40. b — Consumed fraction = 0.40 × 1620 = 648 kJ m■² yr■¹.

41. c — NPP = 3350–855=2495; NPP:GPP = 2495/3350 = 0.74.

42. d — 31,000 × 0.10 × 0.20 = 620 kJ.

43. c — Respiration = GPP – NPP = 6.30 – 4.72 = 1.58.

44. b — Energy transferred = 50,000 × 12/100 = 6,000 kJ.

45. c — NPP = GPP – R = 5300 – 760 = 4540 g C m■² yr■¹.

46. d — Three transfers: 106,000 × 0.1^3 = 106.0 kJ.

47. d — R = GPP – NPP = 5440 – 2540 = 2900.

48. c — 6.4 ha × 10,000 m²/ha × 1600 g/m² = 102,400 kg C.

49. c — Assimilation efficiency = (assimilated – egested)/ingested × 100 = (2600–650.0)/2600 ×100 = 75.0%.

50. d — Consumed fraction = 0.40 × 1820 = 728 kJ m■² yr■¹.

51. b — Standing crop measures living material (often biomass); standing state measures the quantity of inorganic nutrients in an ecosystem.

52. a — NPP = GPP – Ra; heterotrophic respiration is not subtracted in this definition.

53. b — Large energy losses as heat and through metabolism constrain the number of trophic levels.

54. c — Biomagnification is trophic-level amplification of persistent contaminants.

55. d — Bioaccumulation is organism-level uptake exceeding elimination; biomagnification is increase across trophic levels.

56. c — Energy diminishes at each trophic transfer; biomass and numbers may be inverted.

57. b — Phytoplankton have low standing biomass but high productivity and rapid turnover.

58. c — Detrital pathways start with dead organic material and are central to decomposition and mineralisation.

59. a — Keystone impact is disproportionate to abundance or biomass.

60. a — Beavers physically modify habitat and resource availability, affecting many species.

61. b — Facilitation describes positive modification of the environment by earlier colonists.

62. a — A sere is the sequence of successional stages from pioneer to later communities.

63. c — Primary succession begins where soil is absent or essentially undeveloped.

64. b — Lichens and microbial communities can colonise bare rock and contribute to weathering and soil formation.

65. d — Modern ecology recognises dynamic, disturbance-dependent vegetation states rather than a single immutable endpoint.

66. c — A niche includes abiotic tolerances, resource use and biotic interactions, not merely location.

67. d — Complete niche overlap for a limiting resource can lead to exclusion unless coexistence mechanisms operate.

68. c — Species can coexist when they use different resources, times or microhabitats.

69. a — Mutualism is a +/+ interaction, although dependence can vary.

70. d — One species benefits while the other experiences no significant effect.

71. a — One species is inhibited or harmed while the other is unaffected.

72. c — Parasites exploit hosts and commonly depend on them for extended periods.

73. a — Type III species produce many offspring, with high early mortality.

74. d — Type II curves show roughly constant per-capita mortality across life.

75. d — Immigration adds individuals; births also increase size.

76. b — With no migration, ΔN = births – deaths.

77. b — In the exponential model, r = b – d.

78. d — For dN/dt = rN(1–N/K), the absolute growth rate peaks at N=K/2.

79. a — At K, births and deaths balance in the simple logistic model.

80. b — Competition, disease and predation can intensify as density rises.

81. d — Weather extremes can affect populations irrespective of density.

82. c — Biotic interactions restrict where a species actually persists.

83. a — Ecosystem boundaries are analytical choices; fluxes commonly cross them.

84. d — Nitrogenase reduces N■ to ammonia/ammonium during fixation.

85. b — Nitrifying microbes oxidise NH■■ first to NO■■ and then NO■■.

86. c — Denitrifiers use oxidised nitrogen as electron acceptors, returning N to the atmosphere.

87. c — Decomposition/mineralisation releases ammonium from organic nitrogen.

88. c — Phosphorus is primarily sedimentary, with no major gaseous phase.

89. b — Phosphorus is often the limiting nutrient in freshwater; excess input stimulates algal growth.

90. b — Microbial decomposition consumes dissolved oxygen and can cause hypoxia or anoxia.

91. b — Net storage reflects the full carbon budget and persistence of stored pools.

92. a — NEP = NPP − heterotrophic respiration = 1,000 − 700 = 300.

93. c — Decomposition responds strongly to microclimate, substrate chemistry and decomposer communities.

94. c — Lignin is chemically resistant and can slow microbial breakdown.

95. c — Warm, wet tropical forests have high year-round productivity; oceans may have large total production due to area.

96. b — Productivity has time units; standing biomass is an amount at a time.

97. a — Trophic transfer efficiency is energy at one level divided by energy at the preceding level.

98. a — Actual efficiencies vary with organism type, food quality and ecosystem conditions.

99. c — Much terrestrial production is channelled through litter and detritus.

100. b — Resilience concerns persistence or recovery of key functions after disturbance.

101. b — Resistance is the magnitude of immediate response; resilience concerns recovery or persistence.

102. b — Energy flows through ecosystems and is dissipated; nutrients cycle, so statement 3 is false.

103. d — The energy available declines at successive trophic levels because transfer is inefficient.

104. b — Rapid phytoplankton turnover allows consumer biomass to exceed producer standing biomass.

105. b — Nitrification is oxidation of ammonium, not nitrate reduction.

106. c — GPP is gross fixation; NPP subtracts autotrophic respiration; NEP subtracts heterotrophic respiration from NPP.

107. b — A single large producer can support numerous consumers, inverting numbers.

108. d — Mature ecosystems often conserve nutrients efficiently but remain open systems.

109. d — Organic N mineralises to ammonium, which can be nitrified and then denitrified.

110. b — Mycorrhizal fungi improve nutrient/water uptake and receive photosynthate.

111. b — Ecotones are transition zones and may show edge effects.

112. c — Edges can differ from habitat interiors in microclimate, interactions and species composition.

113. c — Diversity indices often combine richness and evenness.

114. b — Dominance by one species reduces evenness while richness remains four.

115. c — Shannon entropy incorporates species proportions and rises as diversity/evenness increase.

116. a — Intermediate disturbance can prevent competitive exclusion without eliminating too many species.

117. b — Bottom-up control originates with resources or primary production.

118. a — Predators indirectly affect lower trophic levels through prey suppression.

119. c — Methylmercury is persistent and efficiently transferred through aquatic food webs.

120. b — Increase within an individual over time is bioaccumulation.

121. d — Persistent, bioavailable, slowly eliminated chemicals can build up and magnify.

122. b — Ecosystem services encompass multiple benefits and ecological processes.

123. c — Anoxic wetland sediments support denitrifiers that convert nitrate to gaseous nitrogen.

124. a — Freshwater systems are often P-limited, though co-limitation and other controls occur.

125. d — Stratification reduces vertical oxygen transfer while decomposition continues.

126. c — Only the energy pyramid is necessarily upright.

127. d — Food webs capture interconnected and often complex feeding relationships.

128. c — Omnivores consume resources from multiple trophic positions.

129. c — Detritivores ingest and fragment detritus; decomposers often chemically break it down externally.

130. a — Consumption, assimilation and production efficiencies have distinct denominators and meanings.

131. b — Energy is lost in egestion; assimilated energy is partitioned into production and respiration.

132. d — Ecological stoichiometry links C:N:P balance to growth, food quality and nutrient cycling.

133. d — Microbes decomposing carbon-rich, N-poor litter may immobilise available nitrogen.

134. a — Biocontrol outcomes depend on specificity, food-web effects and establishment.

135. a — Water limitation constrains desert primary production.

136. c — Estuarine nutrient supply and shallow, dynamic conditions can support high productivity.

137. d — Low areal rates over vast areas can yield substantial total production.

138. c — Upwelling supplies nitrate, phosphate and other nutrients to the euphotic zone.

139. b — The euphotic zone is the illuminated layer where photosynthesis can occur.

140. a — NPP is partitioned among structural tissues, roots, reproduction and herbivory.

141. b — The same biomass stock can arise from very different production and turnover rates.

142. a — Positive NEP means NPP exceeds heterotrophic respiration, implying net ecosystem carbon accumulation before other lateral/disturbance fluxes.

143. c — Thaw-related greenhouse gas release can amplify warming.

144. b — Feedbacks can maintain contrasting ecosystem states, such as clear and turbid lakes.

145. b — Soil carbon accumulates when inputs exceed respiration, erosion and other losses.

146. a — Restoration is context-specific and constrained by altered climate, land use and legacies.

147. a — Biological legacies and intact soil enable faster recolonisation.

148. c — Invasiveness requires establishment and spread with consequential effects; many introduced species do not become invasive.

149. c — Eurytopic species tolerate a broad range; stenotopic species have narrow tolerance.

150. a — The scarcest essential resource relative to organismal requirement can constrain growth.

151. c — Organisms have minimum, optimum and maximum tolerance limits.

152. b — NPP = 2,800–1,100=1,700; NEP = 1,700–1,300=400.

153. d — Assimilated = 1,000–300=700; production = 700–500=200 kJ.

154. b — (242–200)/200 = 0.21 per year.

155. d — $rN(1–N/K)=0.4\times500\times0.5=100$.

156. b — $20{,}000\times0.15\times0.08 = 240$ kJ.

157. c — Ra=1,440; 1,440/3,600×100=40%.

158. c — BCF × water concentration = 500 × 0.004 = 2 mg/kg.

159. d — Strong dominance reduces evenness and therefore lowers many diversity indices.

160. a — Biological production and export of organic matter move carbon to deeper waters.

161. b — Energy pyramids represent fluxes over time, avoiding misleading stock-only comparisons.

162. a — Soil and biological legacies remain, so recovery is secondary succession.

163. c — Bare substrate requires weathering, colonisation and soil formation.

164. a — Oxygen availability strongly affects decomposition pathways and rates.

165. c — Anaerobic metabolism continues; reducing conditions can release phosphorus from sediments.

166. d — Nutrient enrichment can drive eutrophication and subsequent oxygen demand.

167. b — Trophic cascades are indirect effects propagated through food-web links.

168. b — Parasite-mediated changes can alter predation and food-web dynamics.

169. a — Functional responses describe per-capita feeding rate as prey density changes.

170. c — Handling time imposes a maximum intake rate.

171. d — Frequent or intense disturbance can maintain early-stage communities.

172. d — Phosphorus commonly moves with eroded particles and runoff.

173. b — Immobilisation transfers inorganic nutrients into microbial organic matter.

174. c — Litter stoichiometry affects microbial nutrient limitation and decomposition.

175. b — Rapid growth and consumption can maintain low standing stock despite high flux.

176. c — Photosynthesis requires light, nutrients and inorganic carbon.

177. c — Diatoms use dissolved silica to form their cell walls.

178. b — Ecological footprint is an accounting measure of human demand on biocapacity.

179. c — Biochemical oxidation consumes oxygen, raising biochemical oxygen demand.

180. b — BOD estimates oxygen consumed by microbial degradation of biodegradable matter.

181. a — Biological responses integrate conditions but may be influenced by multiple factors.

182. a — Endemism describes restriction to a particular region.

183. c — Larger islands generally have lower extinction rates; less isolated islands receive more immigrants.

184. b — Taking logs gives log S = log c + z log A; z is the slope.

185. c — Metapopulations comprise local populations linked by movement, with local extinction and recolonisation.

186. d — The storage effect allows species to benefit from favourable periods while persisting through unfavourable ones.

187. d — Stability is multidimensional and cannot be reduced to a single metric.

188. c — Multiple attributes of disturbance determine its ecological effects.

189. c — Fire is a characteristic ecological process in many savanna and grassland systems.

190. a — Organic matter accumulation, weathering and biota modify soil through succession.

191. b — Habitat is the place; niche encompasses resource use, tolerances and interactions.

192. b — Density-dependent factors can regulate populations by increasing mortality or reducing reproduction as density rises.

193. b — r = b – d = 0.32 – 0.47 = –0.15.

194. b — K is an environmental property that can vary with conditions.

195. a — Life-history traits vary along continua and depend on ecological context.

196. a — Added N can raise production and favour species able to exploit enriched conditions.

197. a — Excess reactive nitrogen can alter nutrient balance, soil acidity and community composition.

198. b — Respiration by autotrophs, heterotrophs and decomposers releases CO■.

199. d — Methanogenesis occurs under anoxic conditions; methanotrophs oxidise CH■ where oxygen is available.

200. d — Photosynthetic oxygen production and oxidative consumption are coupled to carbon cycling.